

Overview of the Company

The practicum engagement was completed at the City Government of Biñan, a local government unit (LGU) in the Province of Laguna responsible for delivering public services and implementing programs that promote the welfare and development of its constituents. As part of its commitment to improving government operations through digital transformation, various offices within the City Government continuously adopt information technology solutions to streamline administrative processes and enhance the delivery of public services. The practicum was conducted under the office responsible for administering the Financial Assistance Program, which provides educational assistance to qualified senior high school students.

During the practicum, the assigned project involved the design and development of a Financial Assistance Program Voucher Management System, a web-based application intended to replace manual voucher processing and improve the management of beneficiary records. The system was developed to support the registration and maintenance of student information, voucher generation, school management, Excel data importation, PDF document generation, signatory management, user administration, audit logging, and record archiving. These functions were designed to improve the efficiency, accuracy, and organization of the voucher processing workflow while minimizing repetitive manual tasks.

The assigned responsibilities throughout the practicum included gathering and analyzing system requirements, designing and normalizing the database structure, developing system modules using the CodeIgniter 4 framework, implementing both front-end and back-end functionalities, integrating third-party libraries for Excel and PDF processing, conducting functional testing and debugging, and preparing the corresponding technical documentation and user manual. Revisions to the system were also implemented based on feedback from supervisors and end users to ensure that the application met the operational requirements of the host training establishment.

Throughout the engagement, a total of 486 hours were completed in accordance with the practicum requirements of the academic institution. The practicum provided valuable opportunities to apply classroom knowledge to a real-world software development project while strengthening technical competencies in web application development, database management, software testing, documentation, and system deployment. Furthermore, the experience fostered the development of professional skills such as communication, teamwork, problem-solving, adaptability, and time management, all of which are essential in the practice of Information Technology.

Mission:

City of Biñan's leadership is dedicated to fulfilling its responsibilities by (City Hall of Biñan, 2026):

- Promote social order and ensure public safety.
- Enhance economic prosperity through job generation, manpower and skills development, encourage and support local industries and its expansion.
- Guarantee social justice by way of ensuring basic services and equal opportunities.
- Deliver accessible and quality healthcare services.
- Protect and maximize the utilization of environment and natural resources towards the attainment of a disaster-resilient community.
- Ensure access to free and quality education to prepare our youth to be globally competitive.
- Support gender sensitivity, responsiveness and equality.
- Provide modern and needed infrastructure facilities.
- Preserve, conserve and nurture its cultural, historical and artistic heritage toward sustainable tourism.
- Increase collection efficiency through reasonable taxes, fees and charges, and ensure sound fiscal management.

Vision:

According to City Hall of Biñan (2026), its vision is "A modern and developed City where its responsible people are proud of their cultural, historical and artistic heritage; enjoy peace and security, economic stability, social justice, preserved environment, accessible quality education, responsive social services, well-planned infrastructure facilities all anchored on good governance thus making the City of Biñan as the Premiere Heritage and Trade Capital of the South, the best place to live, work and visit."

Reference: City Hall of Biñan, (2026), Mission & Vision, <https://www.binan.gov.ph/mission-and-vision/>

System Development Overview

1. Discussion of the Nature of Tasks

The primary objective of the developed system is to facilitate the efficient administration of the Financial Assistance Program implemented by the City Government of Biñan. The system digitizes the traditional voucher processing workflow by integrating various administrative functions into a centralized web-based application. Specifically, the system supports the management of beneficiary records, participating schools, voucher generation, document printing, data importation, user administration, signatory management, audit logging, and record archiving.

Throughout the practicum, the assigned tasks encompassed the analysis of existing manual procedures, database design and normalization, development of application modules, implementation of business logic, system testing, debugging, and documentation. Additional responsibilities included integrating Excel import functionality, implementing PDF voucher generation, maintaining system security through role-based authentication, and revising the database structure based on user feedback and system requirements. These activities collectively contributed to improving the efficiency, accuracy, and maintainability of the voucher management process.

Table 1. Major Functional Tasks of the System

Functional Area	Description
Student Management	Maintains beneficiary records and eligibility information.
School Management	Stores participating schools and related information.
Voucher Generation	Generates financial assistance vouchers and printable documents.
Excel Import	Imports beneficiary records from standardized Excel templates.
Archive Management	Archives inactive or previous school year records while preserving historical data.
User Management	Manages administrator accounts and system access privileges.
Signatory Management	Maintains authorized signatories used in voucher generation.
Audit Logging	Records significant user activities performed within the system.

PDF Processing	Generates printable voucher documents for distribution.
----------------	---

2. Software Used

The Financial Assistance Program Voucher Management System was developed using a web-based software architecture based on the **Model-View-Controller (MVC)** framework. The application was implemented using **PHP** and the **CodeIgniter 4 Framework**, which provides a structured environment for developing scalable and maintainable web applications.

The system utilizes **MySQL** as its relational database management system for storing and retrieving application data. The user interface was developed using **Bootstrap 5**, **HTML5**, **CSS3**, **JavaScript**, **AJAX**, and **jQuery** to provide a responsive and user-friendly experience.

Several third-party libraries were integrated into the system to enhance functionality. **PhpSpreadsheet** was utilized to facilitate Excel data importation, while **mPDF** was incorporated to support automatic generation of printable voucher documents in Portable Document Format (PDF). System dependency management was handled using **Composer**, whereas **Git** served as the version control system throughout the software development lifecycle.

Table 2. Software Components Used

Software	Purpose
PHP 8.x	Server-side programming language
CodeIgniter 4	MVC web application framework
MySQL	Relational Database Management System
Bootstrap 5	Responsive user interface framework
HTML5 / CSS3	Front-end development
JavaScript / jQuery / AJAX	Client-side interactivity
PhpSpreadsheet	Excel file import processing
mPDF	PDF voucher generation
Composer	Dependency management
Git	Version control system

XAMPP	Local web server environment
-------	------------------------------

3. Hardware Used

The developed system follows a client-server architecture. The server environment requires a computer capable of hosting Apache, PHP, and MySQL services through XAMPP or an equivalent web server package. Client devices require only a modern web browser and network connectivity to access the application.

The recommended hardware specifications are presented below.

Table 3. Recommended Hardware Specifications

Component	Minimum Specification
Processor	Intel Core i3 or AMD Ryzen 3
Memory	8 GB RAM
Storage	256 GB SSD
Operating System	Windows 10/11 (64-bit)
Network	Local Area Network (LAN) or Internet Connection
Display	1366 × 768 Resolution

The recommended specifications ensure efficient execution of database transactions, PDF generation, and Excel file processing, particularly when handling large volumes of beneficiary records.

4. Software Development Outputs

The primary output of the practicum is a fully functional web-based Voucher Management System capable of supporting the operational requirements of the Financial Assistance Program. The developed software integrates multiple administrative modules into a unified application to minimize manual processing and improve operational efficiency.

The project also produced several supporting deliverables, including a normalized relational database, user documentation, system installation manual, technical documentation, and source code repository.

Table 4. Software Development Outputs

Output	Description
Voucher Management System	Web-based application for voucher processing
Database Schema	Normalized MySQL database
Student Management Module	Maintains beneficiary information
School Management Module	Maintains participating schools
User Management Module	Controls system access
Signatory Module	Maintains authorized signatories
Archive Module	Stores historical records
Audit Log Module	Records user activities
Excel Import Module	Imports student information
PDF Generation Module	Produces printable vouchers
System Documentation	User Manual and Technical Documentation

5. Scope of the Project

The developed system is intended to automate the administrative processes associated with the Financial Assistance Program implemented by the City Government of Biñan. Specifically, the project focuses on the management of beneficiary records, participating schools, voucher generation, document printing, Excel-based data importation, archive management, and user administration.

The system supports role-based access control to ensure that only authorized personnel can access administrative functions. Furthermore, audit logging mechanisms provide accountability by recording significant user activities performed within the application.

However, the project is limited to the operational requirements of voucher administration. Financial accounting, budgeting, payroll processing, online payment integration, and external government information systems are outside the scope of the present implementation.

Table 5. Scope and Limitations

Included	Excluded
Student Management	Accounting System
School Management	Payroll Management
Voucher Generation	Online Payment Processing
PDF Generation	Financial Reporting
User Management	Inventory Management
Excel Import	Mobile Application
Audit Logs	SMS Notification System
Archive Management	Integration with External Government Databases

6. Screens of the Features

The system consists of multiple functional modules that collectively support the complete voucher management workflow. Each module provides a dedicated interface designed to simplify administrative tasks while maintaining consistency throughout the application.

The following screens illustrate the primary features of the developed system.

Table 6. System Screens

Screen	Purpose
Login	User authentication
Dashboard	Summary of system information
Student Management	Student registration and maintenance
School Management	School information management
Voucher Generation	Voucher creation and printing
Excel Import	Import beneficiary records
Signatory Management	Maintain authorized signatories
Archive	Archived student records

User Management	Administrator account management
Audit Logs	System activity monitoring
Profile	User profile management

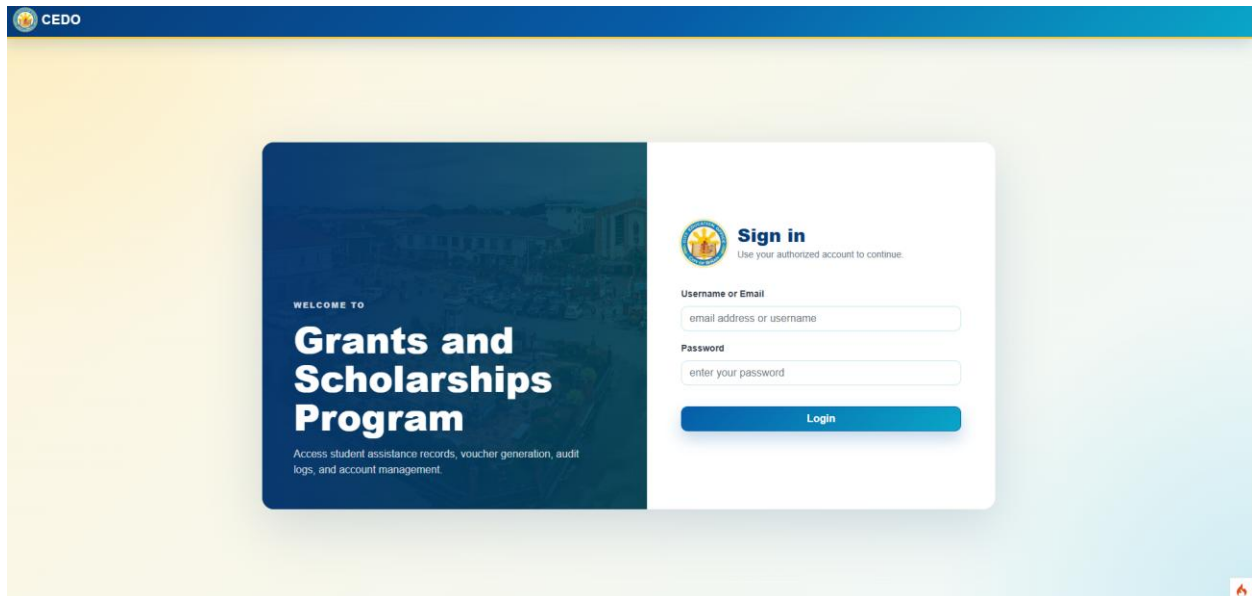


Figure 6.1. Login Page

Figure 6.1 presents the Login Screen of the Financial Assistance Program Voucher Management System. This interface serves as the system's entry point, requiring users to authenticate using their registered username and password before accessing any system functions.

The login module helps ensure that only authorized personnel can access the application, thereby protecting sensitive student information and administrative records.

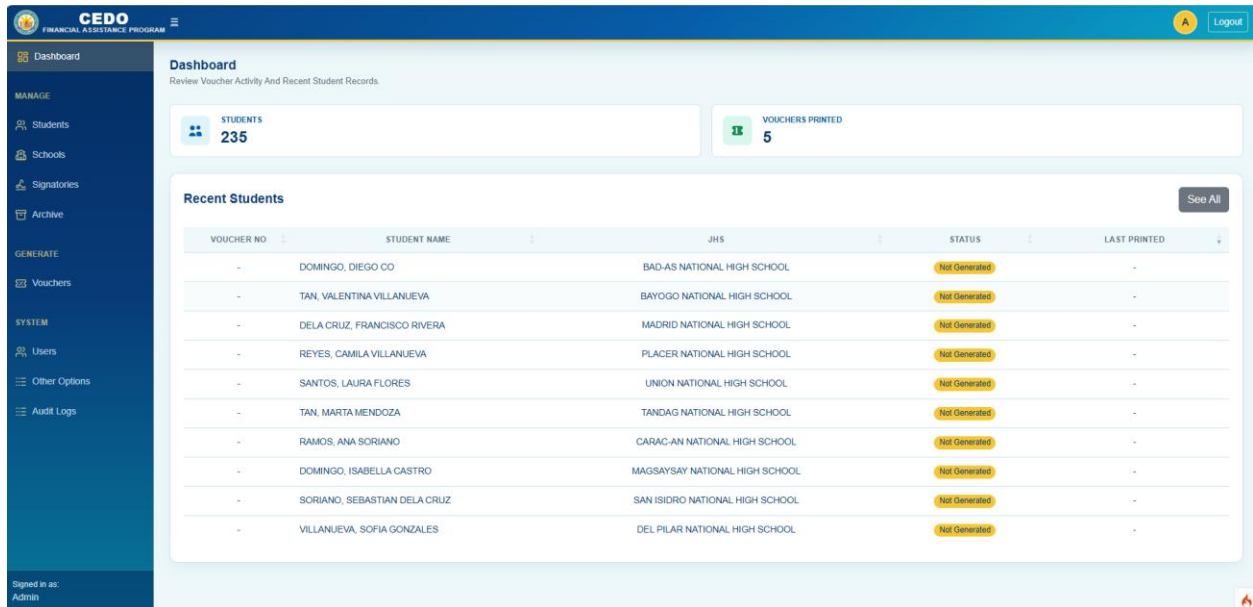


Figure 6.2. Admin Dashboard

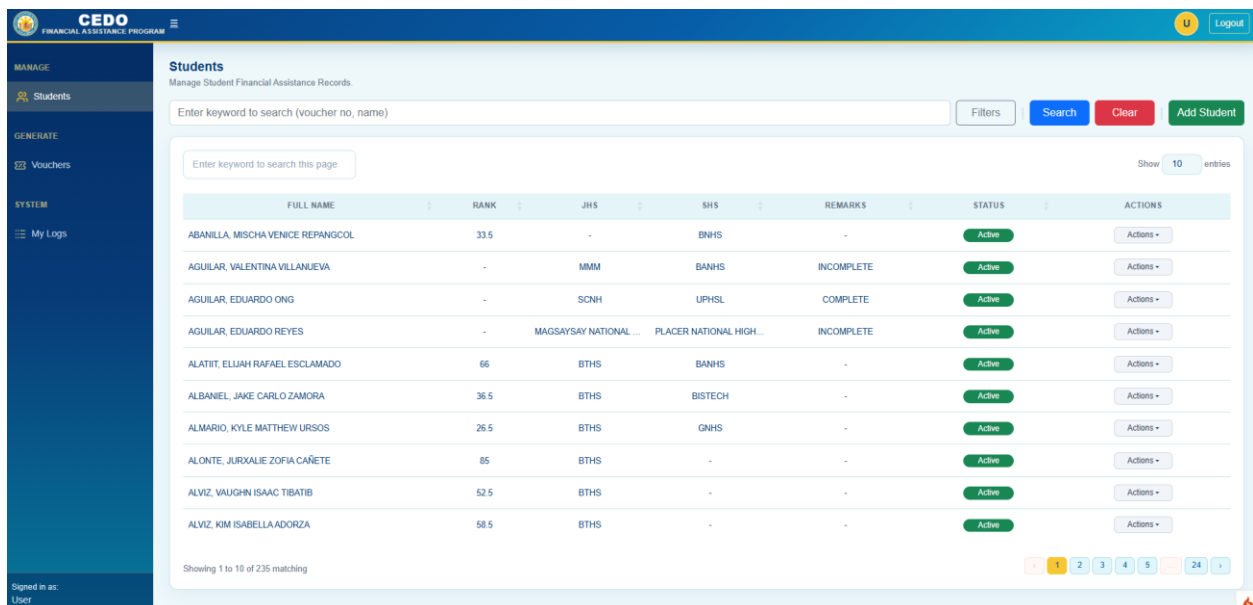


Figure 6.3. User Dashboard

Figure 6.2 and 6.3 shows the Dashboard, which serves as the central landing page after successful authentication. It provides users with an overview of the system and quick access to the major modules of the application.

The dashboard improves navigation by allowing administrators to access frequently used functions from a single interface.

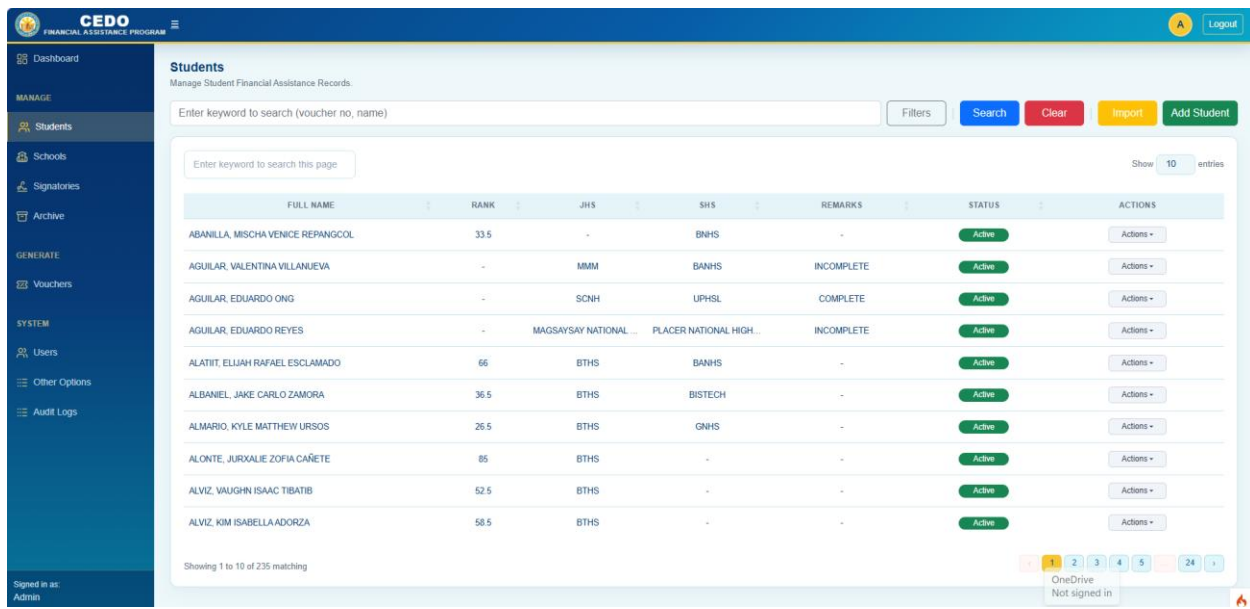


Figure 6.4. Student Management Page

Figure 6.4 illustrates the Student Management module. This module displays the list of registered beneficiaries and allows authorized users to add, edit, archive, and search student records.

It serves as the primary repository of beneficiary information used throughout the voucher generation process.

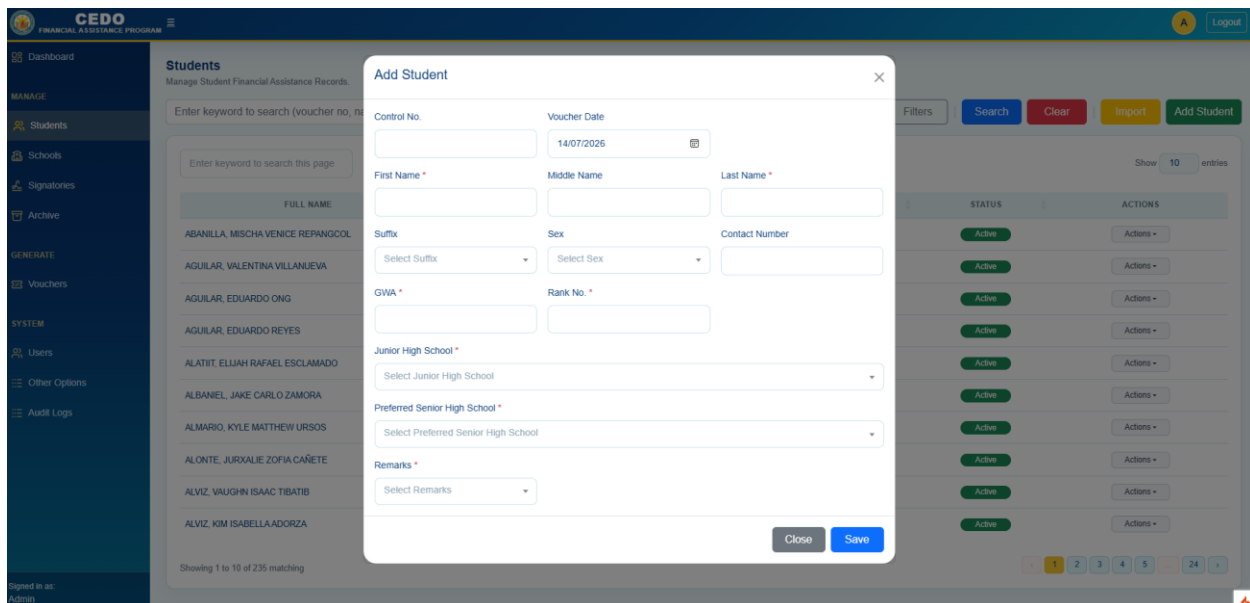


Figure 6.5. Add Student Modal

Figure 6.5 presents the Add Student interface, where users manually encode the personal and academic information of financial assistance beneficiaries.

The system validates the submitted information before saving the record to ensure completeness and consistency of stored data.

The screenshot shows the 'Edit Student Modal' in the CEDO Financial Assistance Program. The modal is open over a table of student records. The modal contains the following fields:

- Control No. (text input)
- Voucher Date (date input, format dd/mm/yyyy)
- First Name (text input, value: MISCHA VENICE)
- Middle Name (text input, value: REPANGCOL)
- Last Name (text input, value: ABANILLA)
- Suffix (dropdown menu, value: Select Suffix)
- Sex (dropdown menu, value: FEMALE)
- Contact Number (text input)
- GWA (text input, value: 95.64)
- Rank No. (text input, value: 33.5)
- Junior High School (dropdown menu, value: Select Junior High School)
- Preferred Senior High School (dropdown menu, value: BNHS - BAYOGO NATIONAL HIGH SCHOOL - SENIOR HIGH)
- Remarks (dropdown menu, value: Select Remarks)

The modal has 'Close' and 'Update' buttons at the bottom right. The background table shows a list of students with columns for Full Name, Rank, JHS, SHS, Remarks, Status, and Actions.

Figure 6.6. Edit Student Modal

Figure 6.6 shows the Edit Student interface. This page allows authorized users to update existing beneficiary records whenever corrections or modifications are necessary.

Any changes made through this module are reflected in the database and may be recorded in the audit log for accountability.

The screenshot shows the 'Excel Import' modal in the CEDO Financial Assistance Program. The modal is open over a table of student records. The modal contains the following fields:

- Upload a .xlsx, .xls, or .csv file.
- File (text input, value: No file chosen)
- Choose File (button)
- Import (button)

The modal has 'Close' and 'Import' buttons at the bottom right. The background table shows a list of students with columns for Full Name, Rank, JHS, SHS, Remarks, Status, and Actions.

Figure 6.7. Excel Import

Figure 6.7 presents the Excel Import module, which enables users to upload multiple student records using a standardized spreadsheet template.

This feature minimizes manual data entry and significantly reduces the time required to register large numbers of beneficiaries.

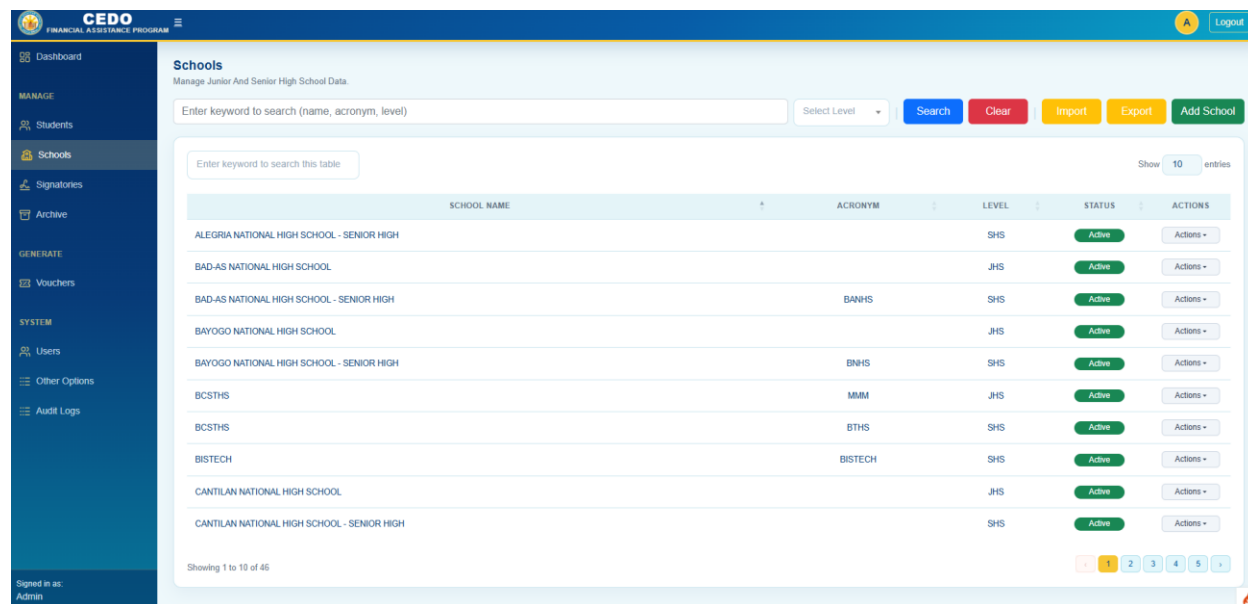


Figure 6.8. School Management Page

Figure 6.8 illustrates the School Management module. This interface allows administrators to maintain the list of participating schools that may be selected during student registration.

Maintaining an updated school list promotes data consistency and reduces repetitive manual encoding.

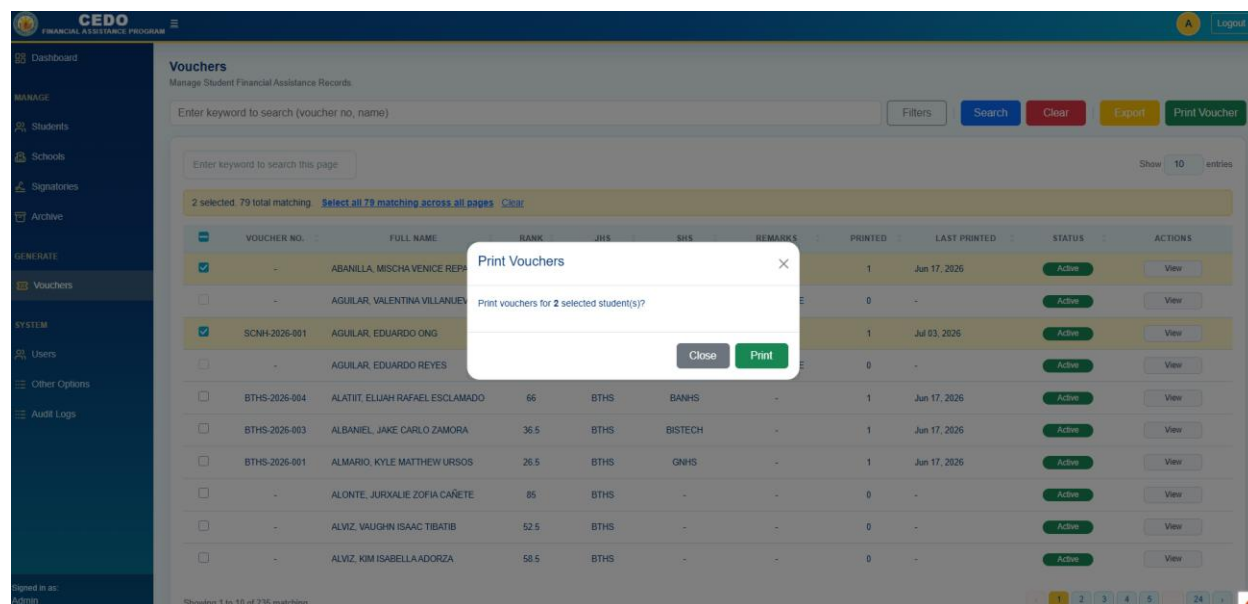


Figure 6.9. Voucher Generation

Figure 6.9 shows the Voucher Generation module. This module automatically prepares financial assistance vouchers using the selected student's information together with the currently active signatories.

The generated voucher may then be reviewed, exported, or printed for official distribution.

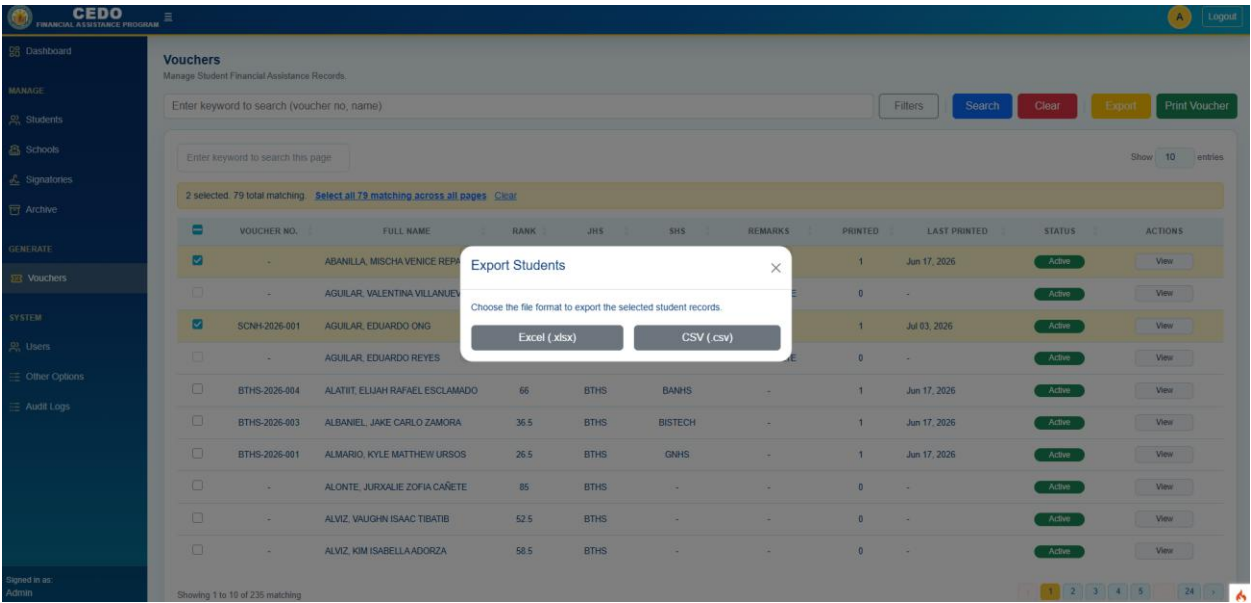


Figure 6.10. PDF Generation

Figure 7 presents the PDF Generation interface. This module converts voucher information into printable PDF documents that preserve the official layout and formatting.

The feature simplifies document preparation and supports efficient voucher distribution.

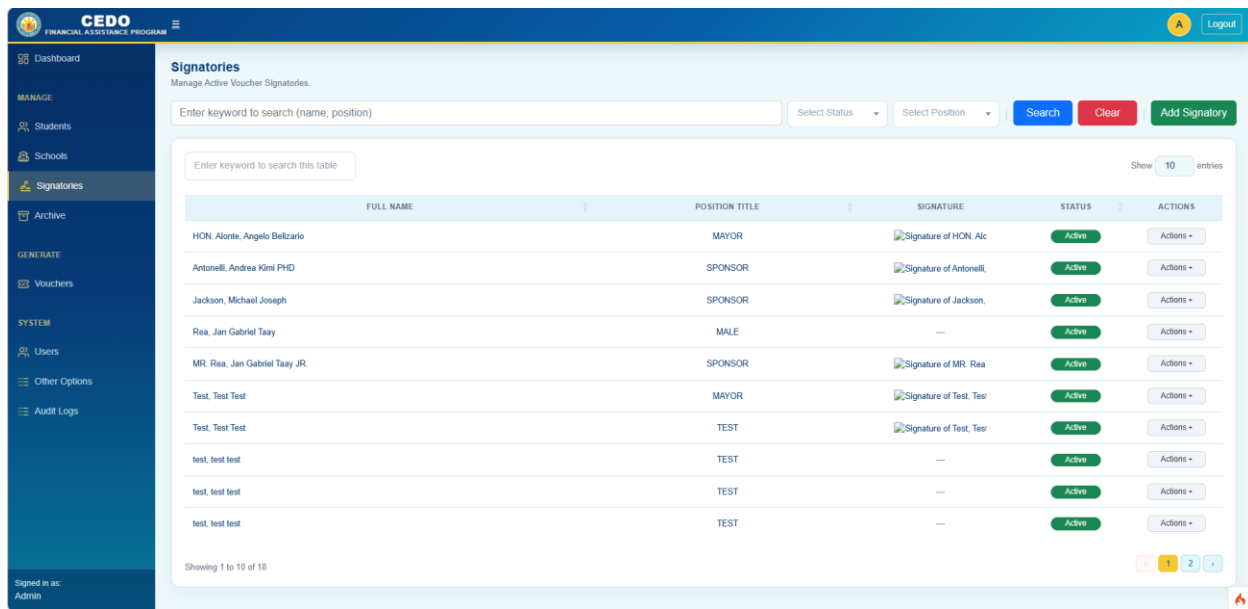


Figure 6.11. Signatories Management Module

Figure 6.11 illustrates the Signatory Management module. This page enables administrators to add, update, activate, or deactivate signatories whose names and signatures appear on generated vouchers.

Changes made within this module are automatically reflected in future voucher generation without requiring modifications to the voucher template.

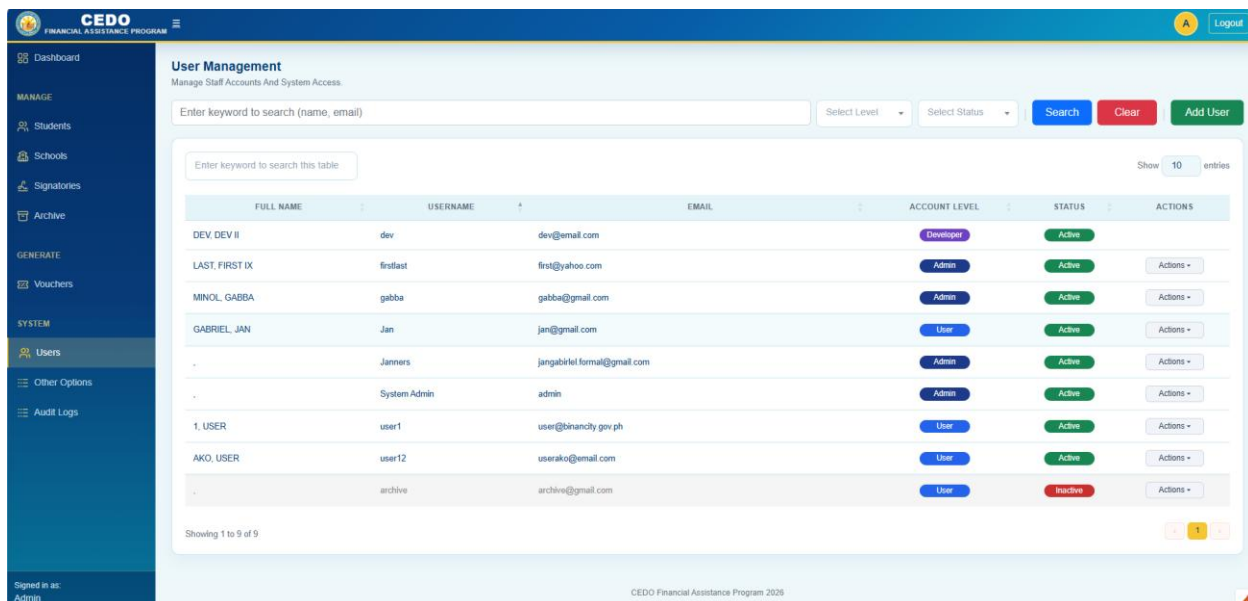


Figure 6.12. User Management Module

Figure 6.12 presents the User Management module. This interface allows administrators to create, update, activate, and manage system user accounts.

Proper user management ensures that only authorized personnel can access the system and perform administrative functions.

Audit Logs
Track Account Activity And Voucher Changes.

Enter keyword to search (action, description, user) Select Action dd/mm/yyyy dd/mm/yyyy Search Clear

Enter keyword to search this table Show 10 entries

DATE/TIME	ACTION	DESCRIPTION	USER AGENT	USER
Jul 14, 2026 02:48 PM	LOGIN	User ADMIN AKO logged in	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2
Jul 14, 2026 02:48 PM	LOGOUT	User USER AKO logged out	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	USER AKO #10
Jul 14, 2026 02:47 PM	LOGIN	User USER AKO logged in	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	USER AKO #10
Jul 14, 2026 02:47 PM	LOGOUT	User ADMIN AKO logged out	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2
Jul 14, 2026 02:46 PM	LOGIN	User ADMIN AKO logged in	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2
Jul 14, 2026 02:46 PM	LOGIN_FAILED	Bad password for "admin@binancity.gov.ph"	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2
Jul 14, 2026 10:10 AM	LOGIN	User ADMIN AKO logged in	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2
Jul 14, 2026 10:10 AM	LOGOUT	User USER AKO logged out	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	USER AKO #10
Jul 14, 2026 10:10 AM	LOGIN	User USER AKO logged in	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	USER AKO #10
Jul 14, 2026 10:09 AM	LOGOUT	User ADMIN AKO logged out	MaciBa5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Ch...	ADMIN AKO #2

Showing 1 to 10 of 1,000

Figure 6.13. Audit Logs

Figure 6.13 shows the Audit Log module. This page records significant user activities, including login attempts, record modifications, voucher generation, and administrative actions.

The audit log promotes accountability by providing administrators with a chronological history of system activities.

Archive
View Archived Student Records. Use Filters To Narrow By School Year.

Enter keyword to search (name, school) Filters Search Clear Archive Current Data Export

Enter keyword to search this page Show 10 entries

VOUCHER NO.	FULL NAME	RANK	JHS	SHS	REMARKS	STATUS	SCHOOL YEAR
BNHS-2026-002	1, 1 11	1	BAYOGO NATIONAL HIGH ...	ALEGRIA NATIONAL HIGH ...	-	Generated	2026-2027
LNHS-2026-002	1, 1 1	1	LINIBUNAN NATIONAL HIGH ...	STA. CRUZ NATIONAL HIG ...	-	Generated	2026-2027
VOC-2026-000001	1, 1	0	-	STA. CRUZ NATIONAL HIG ...	-	Generated	2026-2027
DPNHS-2026-001	22, 22 22	2	DEL PILAR NATIONAL HIG ...	ALEGRIA NATIONAL HIGH ...	COMPLETE	Generated	2026-2027
MNHS-2026-001	AAAA, JAN GABRIEL TAAY	-	MADRID NATIONAL HIGH ...	MADRID NATIONAL HIGH ...	-	Generated	2026-2027
VCH-2025-9122	ABADILLA, TRANQUILINO ...	122	-	-	COMPLETE	Generated	2026-2027
VCH-2025-9122	ABADILLA, TRANQUILINO ...	122	LUPAC NATIONAL HIGH S...	MOGPOG SHS	COMPLETE	Generated	2025-2026
-	ABANILLA, MISCHA VENIC...	33.5	BCSTHS	-	-	Not Generated	2026-2027
PNHS-2026-001	AEA, JAN GABRIEL TAAY	-	PANAYOGAN NATIONAL H...	MAGSAYSAY NATIONAL HI...	-	Generated	2026-2027
VOC-2025-000129	AGUILAR, RONNIE DELA ...	29	-	CLAVER NATIONAL HIGH ...	INCOMPLETE	Generated	2026-2027

Showing 1 to 10 of 674 matching

Figure 6.14. Archive Page

Figure 6.14 illustrates the Archive module, which stores beneficiary records that are no longer part of the active school year or have already completed the voucher processing cycle.

Archiving records preserves historical information while maintaining an organized and uncluttered active database.

The screenshot displays the CEDO Financial Assistance Program interface. A 'My Account' modal is open, allowing users to update their profile information and change their password. The background shows the 'Archive' module with a table of archived student records.

My Account

Profile Information
This information appears in logs and account activity

USERNAME: admin1 **EMAIL:** admin@binacity.gov.ph

FIRST NAME: ADMIN **MIDDLE NAME:**

LAST NAME: AKO **SUFFIX:** Select Suffix

Change Password
Leave these fields blank if you do not want to change your password.

CURRENT PASSWORD:

NEW PASSWORD: **CONFIRM PASSWORD:**

Archive Module Table:

VOUCHER NO.	STATUS	SCHOOL YEAR
BNHS-2026-002	1, 1, 11	
LNHS-2026-002	1, 1, 1	
VOC-2026-000001	1, 1	
DPNHS-2026-001	22, 22, 22	
MNHS-2026-001	AAAA, JA	
VCH-2025-0122	ABADILLA	
VCH-2025-0122	ABADILLA	
	ABANILLA	
PNHS-2026-001	AEA, JAN	
VOC-2025-000129	AGUILAR	

Figure 6.15. Profile Settings

Figure 6.15 presents the Profile Settings page, where users can review and update their personal account information.

This module enables users to maintain accurate account details while supporting secure account management.




BIÑAN CITY GRANTS AND SCHOLARSHIP PROGRAMS
FINANCIAL ASSISTANCE PROGRAM
CITY ORDINANCE NO. 08-(2022)

Voucher No.: VCH-2025-0001 Date: 03/10/2025
Name of Recipient: Juan Santos Dela Cruz
Senior High School: MARINDUQUE STATE COLLEGE SHS
Amount in words: TEN THOUSAND PESOS ONLY **PHP 10,000.00**

MR. JAN GABRIEL T. REA JR.
Sponsor




BIÑAN CITY GRANTS AND SCHOLARSHIP PROGRAMS
FINANCIAL ASSISTANCE PROGRAM
CITY ORDINANCE NO. 08-(2022)

Voucher No.: VCH-2025-0002 Date: 03/10/2025
Name of Recipient: Maria Reyes Garcia
Senior High School: MARINDUQUE NATIONAL HIGH SCHOOL SHS
Amount in words: TEN THOUSAND PESOS ONLY **PHP 10,000.00**

MR. JAN GABRIEL T. REA JR.
Sponsor




BIÑAN CITY GRANTS AND SCHOLARSHIP PROGRAMS
FINANCIAL ASSISTANCE PROGRAM
CITY ORDINANCE NO. 08-(2022)

Voucher No.: VCH-2025-0004 Date: 03/11/2025
Name of Recipient: Ana Cruz Torres
Senior High School: MARINDUQUE STATE COLLEGE SHS
Amount in words: TEN THOUSAND PESOS ONLY **PHP 10,000.00**

Figure 6.16. Generated Voucher

Figure 6.16 displays a sample generated financial assistance voucher. The document contains the beneficiary's information, voucher details, and the currently active signatories required for official approval.

The standardized voucher format ensures consistency and simplifies the printing and distribution process.

7. Discussion

The developed Financial Assistance Program Voucher Management System successfully demonstrates the application of web-based information systems in improving administrative efficiency within a government environment. By replacing manual voucher processing procedures with a centralized digital platform, the system minimizes redundant data entry, improves record accuracy, and enhances document generation.

Furthermore, the implementation of role-based authentication, audit logging, and record archiving promotes accountability, data integrity, and long-term maintainability. The integration of Excel import functionality, and automated PDF generation significantly reduces processing time, enabling administrators to manage beneficiary information more efficiently.

Overall, the developed system addresses the operational requirements identified by the host training establishment and provides a scalable foundation for future enhancements. Its modular architecture also facilitates future maintenance and expansion as organizational requirements evolve.

Synthesis of the Practicum Engagement

The practicum engagement at the City Government of Biñan provided an opportunity to apply the theoretical knowledge acquired throughout the Information Technology program in a real-world working environment. Participating in the development of the Financial

Assistance Program Voucher Management System allowed the integration of software engineering principles, database management, web development, and system analysis into a practical project that addressed actual administrative needs. Through the development process, technical skills in PHP, CodeIgniter 4, MySQL, JavaScript, Bootstrap, and version control were further strengthened while gaining experience in designing scalable and maintainable information systems.

Beyond technical competencies, the practicum emphasized the importance of communication, collaboration, and adaptability in a professional setting. System requirements evolved throughout the development process as a result of consultations with supervisors and end users, requiring revisions to the database design, system modules, and documentation. These experiences highlighted that software development is an iterative process where continuous feedback, testing, and refinement are essential in producing solutions that effectively address organizational requirements. Managing changing requirements also reinforced the importance of planning, time management, and maintaining organized documentation throughout the project lifecycle.

The practicum also provided valuable insights into the role of information technology within the public sector. Developing a system for a government office demonstrated how digital solutions can improve operational efficiency by reducing manual processing, minimizing errors, and enhancing record management. The implementation of features such as audit logging, role-based access control, record archiving, and automated voucher generation illustrated the significance of building systems that prioritize data integrity, accountability, and usability alongside functionality.

Overall, the practicum served as a meaningful bridge between academic learning and professional practice. It enhanced both technical and interpersonal competencies while fostering a deeper appreciation for the responsibilities involved in developing information systems for real users. The experience strengthened confidence in applying software development methodologies, solving practical problems, and collaborating with stakeholders to produce solutions that support organizational objectives. More importantly, it reinforced the value of continuous learning and professional growth, preparing the trainee for future roles in the field of Information Technology.

Appendices